# Chapter 12, Problem 4P

## Problem

A three-phase system with *abc* sequence and $VL$ = 440 V feeds a Y-connected load with $Z_L = 40\angle 30°\ \Omega$ . Find the line currents.

## Step-by-step solution

**Step 1** of 2

The line voltage is, $\mathbf{V}_L = 440\ \text{V}$

Calculate the phase voltage.

$$\begin{aligned}\mathbf{V}_p &= \frac{\mathbf{V}_L}{\sqrt{3}}\\ &= \frac{440}{\sqrt{3}}\\ &= 254\ \text{V}\end{aligned}$$

The phase sequence is $abc$.

Therefore,

$$\begin{aligned}\mathbf{V}_{an} &= \mathbf{V}_p\angle 0°\\ &= 254\angle 0°\ \text{V}\\ \mathbf{V}_{bn} &= \mathbf{V}_p\angle -120°\\ &= 254\angle -120°\ \text{V}\\ \mathbf{V}_{cn} &= \mathbf{V}_p\angle 120°\\ &= 254\angle 120°\ \text{V}\end{aligned}$$

**Step 2** of 2

Calculate the line currents.

$$\mathbf{I}_a = \frac{\mathbf{V}_{an}}{\mathbf{Z}_L}$$

Substitute $254\angle 0°\ \text{V}$ for $\mathbf{V}_{an}$ and $40\angle 30°$ for $\mathbf{Z}_L$.

$$\begin{aligned}\mathbf{I}_a &= \frac{254\angle 0°}{40\angle 30°} \\ &= 6.35\angle -30°\ \text{A}\end{aligned}$$

$$\begin{aligned}\mathbf{I}_b &= \frac{\mathbf{V}_{bn}}{\mathbf{Z}_L} \\ &= \frac{254\angle -120°}{40\angle 30°} \\ &= 6.35\angle -150°\ \text{A}\end{aligned}$$

$$\begin{aligned}\mathbf{I}_c &= \frac{\mathbf{V}_{cn}}{\mathbf{Z}_L} \\ &= \frac{254\angle 120°}{40\angle 30°} \\ &= 6.35\angle 90°\ \text{A}\end{aligned}$$

Therefore, the line currents are,

$$\boxed{\begin{aligned}\mathbf{I}_a &= 6.35\angle -30°\ \text{A} \\ \mathbf{I}_b &= 6.35\angle -150°\ \text{A} \\ \mathbf{I}_c &= 6.35\angle 90°\ \text{A}\end{aligned}}.$$